VICTORY 2027

Conventional Sources Of Energy

LECTURE-1 Geography

Sameer Sir



Topics To Be Covered

1. Coal, Petroleum
2.
3.
4.

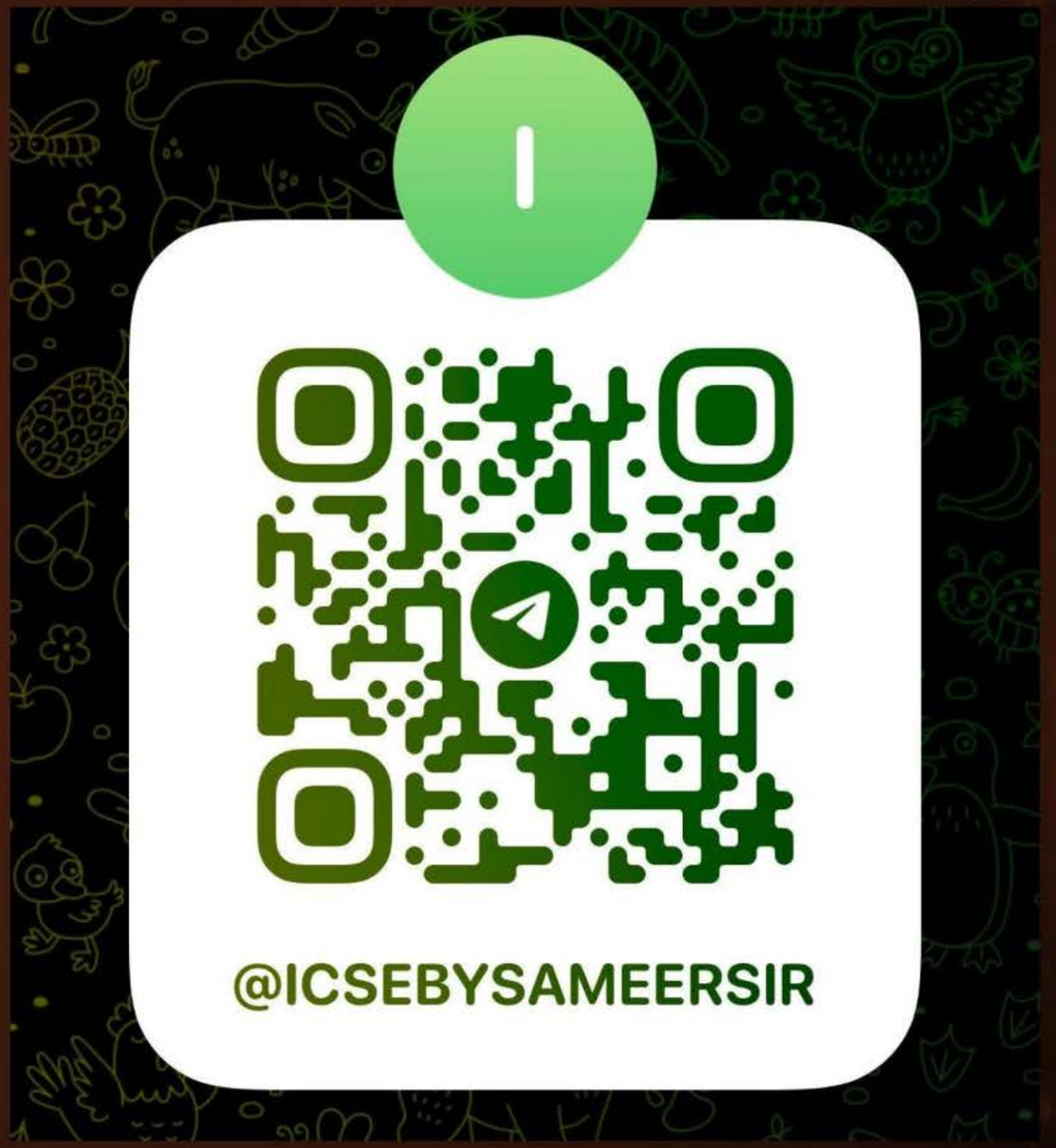
Mineral and Energy Resources

- *Iron ore, Manganese, Copper, Bauxite – uses and their distribution. (two leading states and their prominent mining areas)*
- *Conventional Sources: Coal, Petroleum, Natural gas (distribution, advantages and disadvantages).*
- *Hydel power - Bhakra Nangal Dam and Hirakud Projects and their benefits.*
- *Non-conventional Sources: Solar, wind, tidal, geo-thermal, nuclear and bio-gas (important areas and advantages and disadvantages).*



Telegram

https://t.me/icsebysameersir



Conventional Sources of Energy

- Coal, Petroleum, Natural Gas
  - ↓
  - Fossil fuel
    - Non-renewable
    - Pollution
    - Costly

Sources of Energy

- Conventional sources of energy are those which have been used by man for a long time and are still being tapped and used abundantly.
- These include **coal, petroleum, natural gas and hydel power.**
- Except for hydel power, these sources of energy are non-renewable, cause pollution and are costlier in the longer run than renewable sources of energy.

chemicalengineeringworld.com



Coal



Crude Oil



Natural Gas

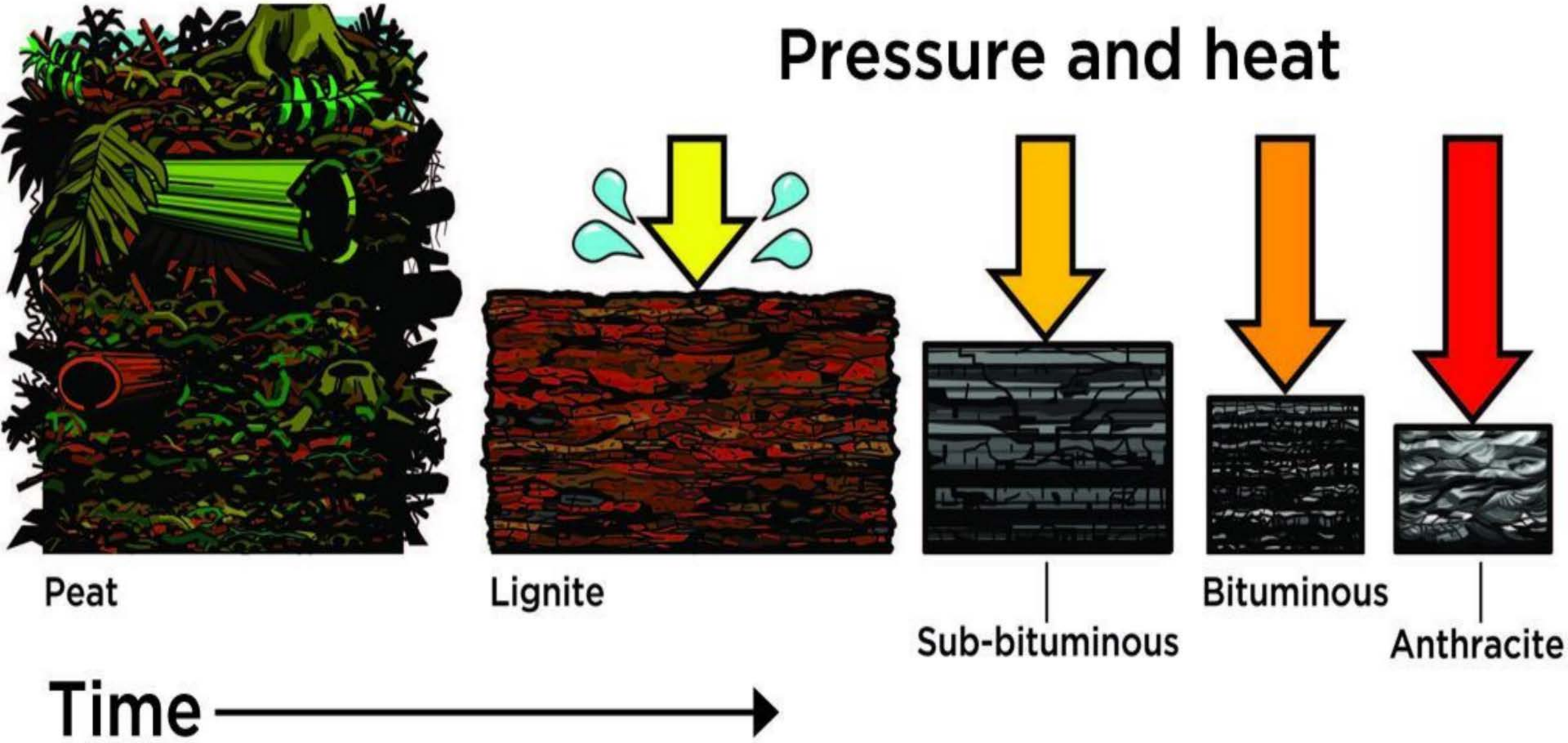


Coal

- **Coal is a combustible solid stratified rock of organic and mineral matter.**

- It is composed of
  - carbon (60 to 90per cent),
  - hydrogen (1 to 12per cent),
  - oxygen (2 to 20 per cent),
  - nitrogen (1 to 3per cent) and also
  - small amounts of phosphorus and sulphur.

- It occurs as a sedimentary rock.


Pressure and heat
Peat
Lignite
Sub-bituminous
Bituminous
Anthracite
Time

Coal

- Coal is formed due to the accumulation of plant matter.
- This vegetative matter was subjected to geological processes that resulted in physical and chemical changes.
- The mass of vegetative matter was probably changed into coal by heat and pressure generated by the increasing weight of overlying sediments and also by the movements of the earth.
- Depending on the amount of carbon, moisture and volatile matter, coal can be classified as follows.

A - Best →

B

L

P → worst → Earliest formed

Peat	Lignite	Bituminous coal	Anthracite
			
0.5 million years ago	100 million years ago	200 million years ago	300 million years ago

The Ranks of Coal

(Lowest to Highest)



1

Peat

Soft, very high moisture,

Very low carbon content.



2

Lignite (Brown Coal)

Low carbon, high moisture, Low heating value.

Use: Electricity Generation.



Cooking Coal

3

Bituminous Coal

Dense, black, high heating value.

Use: Electricity Generation, *Steelmaking (Coke).*



4

Anthracite

Highesst rank., Hard, shiny, highest carbon, Lowest moisture.

Use: Residential / Commercial Heating. Difficult to ignite.

Coal

(1) Anthracite Coal: It is the hardest and it ranks highest among different varieties of coal.

- It has a carbon content of 90per cent and burns slowly without smoke. It burns for a long time and leaves very little ash behind.

Coal

India

(2) **Bituminous Coal:** It is hard, black and compact and makes up almost 80per cent of the world's total coal output.

- It varies greatly in composition and in carbon content, i.e., from 50 to 80per cent.
- It has the widest utilisation for domestic purposes in India as it is easily available.

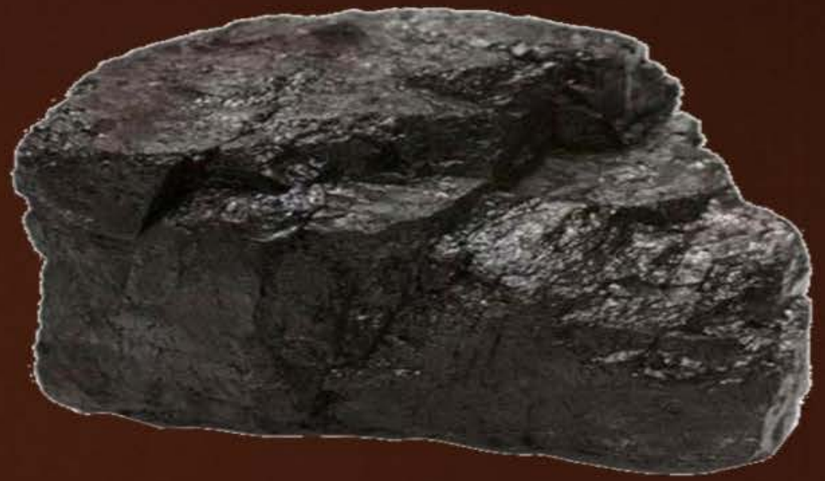
Metallurgical Coal

Coking coal is high-grade bituminous coal which is an important ingredient in iron and steel smelting in blast furnaces. - भट्ठी

पिघलना



Bituminous Coal



Lignite Coal

Coal

(3) **Lignite:** It is also known as brown coal and is a lower grade coal.

- It comprises 40per cent of carbon and a good deal of moisture and less of combustible matter.

Coal

1st form of Coal

(4) **Peat:** It represents the first stage of transformation of wood into coal.

- It has the least carbon content and is inferior to the other three varieties of coal.

Distribution

- India has a total coal reserve of 344.02 billion tonnes and is the second largest producer of coal in the world after China.
- 72%* of the electricity in India is generated from coal.
- **The oldest coalfield in India is in Raniganj in West Bengal and the largest coalfield is in Jharia in Jharkhand.**

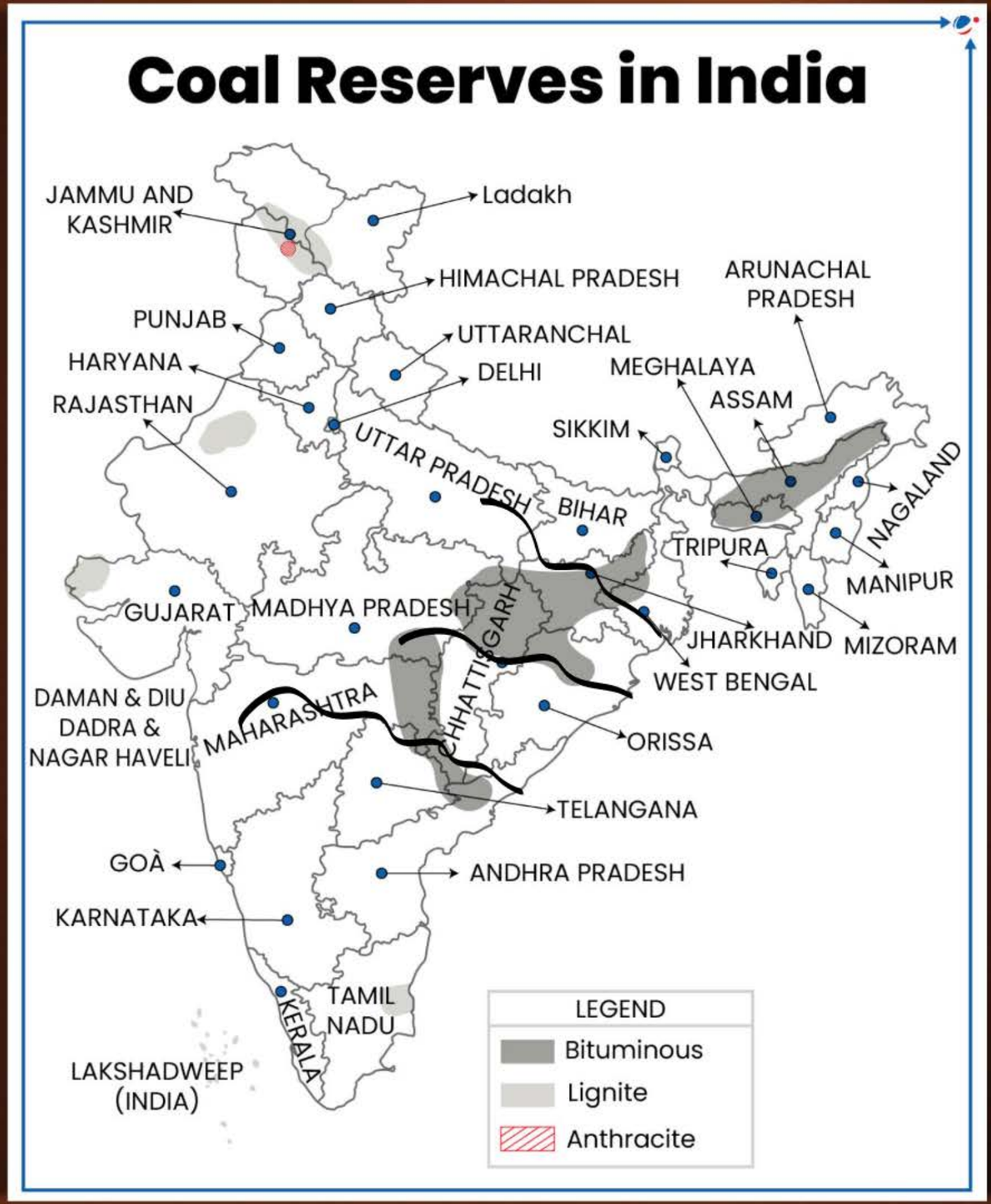
Distribution

- In India, coal occurs in rock sequences mainly of two geological ages, namely,
- **Gondwana, little over 200 million years in age and**
- **Tertiary deposits which were found at a much later geological epoch, i.e., about 55 million years ago.**

- India does not have extensive deposits of the highest grade (90% carbon) anthracite coal.

- The coal of the Gondwana fields, that is, mainly of bituminous quality (50% to 80% of carbon) is mostly available in India.

Feature	Gondwana Coal	Tertiary Coal
Age	A little over 200 million years	About 55 million years
Share	About 4/5 (80%) of India's coal reserves	Small share
Quality	Mainly bituminous, better	More moisture, more sulphur, lower calorific value
Location	River valleys: Damodar, Mahanadi, Son, Godavari	Assam, Arunachal Pradesh, Meghalaya, Nagaland, Kashmir


Coal Reserves in India
JAMMU AND KASHMIR
Ladakh
HIMACHAL PRADESH
ARUNACHAL PRADESH
PUNJAB
UTTARANCHAL
HARYANA
DELHI
MEGHALAYA
ASSAM
RAJASTHAN
SIKKIM
UTTAR PRADESH
BIHAR
NAGALAND
TRIPURA
MANIPUR
GUJARAT
MADHYA PRADESH
CHHATTISGARH
JHARKHAND
MIZORAM
WEST BENGAL
DAMAN & DIU
DADRA & NAGAR HAVELI
MAHARASHTRA
ORISSA
TELANGANA
GOÀ
ANDHRA PRADESH
KARNATAKA
TAMIL NADU
KERALA
LAKSHADWEEP (INDIA)
LEGEND
Bituminous
Lignite
Anthracite

Gondwana → Jharkhand
→ West Bengal
→ Chhatisgarh
→ M.P.
→ Orrisa
→ Maharashtra
→ Telangana
→ Tamil Nadu
→ Gujarat.

Tertiary → Assam
→ meghalaya
→ Kashmir
→ Rajasthan



Distribution and Uses of Different Varieties of Coal in India

Rank	Type	Carbon	Features	Uses	Where in India
4 (lowest)	Peat	Least (<40%)	Soft, very high moisture, first stage	Fuel, filter, plant substrate, cattle bedding	Nilgiris, Kashmir valley, swampy coasts
3	Lignite (brown coal)	About 40%	High moisture, low heating value	Thermal electricity	Neyveli (Tamil Nadu), Rajasthan, Gujarat, Kerala, Puducherry
2	Bituminous	About 50-80%	Black, hard, compact, high heating value	Industry, coke for iron and steel (coking coal), electricity	Gondwana fields (Jharkhand, WB, Odisha, Chhattisgarh)
1 (highest)	Anthracite	90% and above	Hardest, shiny, burns slowly, smokeless, very little ash	Metallurgy, domestic/commercial heating	Only small amounts in Jammu and Kashmir

Gondwana coalfields

- Gondwana coalfields are largely confined to the river valleys like those of Damodar, Mahanadi and Godavari.
- The major resources of Gondwana coal are located in coalfields occupying the Indian heartland in the States of West Bengal, Jharkhand, Odisha, Chhattisgarh, Madhya Pradesh, Maharashtra, Uttar Pradesh Andhra Pradesh and Telangana.
- These coalfields account for four-fifths of India's coal reserves.

Gondwana Coalfields

State	River Valley	Coalfields
1. Jharkhand	Damodar	Jharia, Karnpura, Ramgarh, East and West Bokaro, Giridih, Auronga, Hutar and Daltonganj.
2. Odisha	Mahanadi	Sambalpur, Sundargarh and Talcher.
3. Madhya Pradesh	Mahanadi/Son	Singrauli, Narsingpur, Chhindwara, Betul.
4. Chhattisgarh	Mahanadi/Son	Sarguja, Sendurgarh, Rampur, Korba and Raigarh.
5. Maharashtra	Godavari	Kamptee, Wardha valley, Chandrapur and Yavatmal.
6. Andhra Pradesh	Godavari	East and West Godavari.
7. Telangana	Godavari	Adilabad, Khammam and Warangal.
8. West Bengal	Damodar	Raniganj, Bardhaman, Bankura, Purulia, Darjeeling and Jalpaiguri.

Tertiary Coalfields

The Tertiary coal of India as compared to the Gondwana coal, has higher moisture content and more sulphur, with less calorific value.

- Tertiary coalfields are found in Assam, Arunachal Pradesh, Meghalaya, Nagaland and Kashmir.
- Besides, brown coal or lignite occurs in coastal areas of Tamil Nadu, Gujarat and inland basin of Rajasthan.
- The Neyveli lignite field in Tamil Nadu is the largest lignite deposit field in South India.

Advantages

The main advantages of coal are the following:

(i) Coal is used as a source of power for running machines, trains, ships.

(ii) Coal is used for manufacturing iron and steel.

(iii) Coal is used as a source of direct heat and energy for domestic purposes, in the potteries and for building materials like cement, burning of bricks and tiles and in iron and brass factories, etc.

(iv) It is used as raw material in Thermal Power Plants.

Disadvantages

Heat generated on burning 1 kg of coal

(i) Calorific value of coal found in India is low.

(ii) Coal reserves are scattered in small amounts in India.

(iii) Cost of production and transportation of coal is quite high.

(iv) There are limited reserves of coal in India.

(v) Large scale pollution is caused at the site of mining and place of use.

Advantages (Uses)	Disadvantages
Runs thermal power plants	Burning causes air pollution (smoke, CO2, SO2)
Makes iron and steel (coke)	Global warming and acid rain
Domestic and industrial heat (bricks, cement, potteries)	Mining causes land damage, dust, mine accidents
Source of coal gas and chemicals	Low calorific value of Indian coal
Fuel for trains, ships (earlier)	Reserves scattered; high mining and transport cost

Quick facts

- Oldest coalfield: Raniganj (West Bengal).
- Largest/most important coking coalfield: Jharia (Jharkhand).
- Largest lignite field: Neyveli (Tamil Nadu).
- India is the 2nd largest coal producer after China.
- About 70% of India's electricity comes from coal (lecture: 72%).

1. Which is the highest grade of coal with the maximum carbon content?

- A) Bituminous
- B) Peat
- C) Lignite
- D) Anthracite

2. Which variety of coal is most widely used in metallurgical industries and thermal power stations in India?

- A) Anthracite
- B) Bituminous
- C) Lignite
- D) Peat

4. Neyveli in Tamil Nadu is famously known for mining which type of coal?

- A) Bituminous
- B) Lignite
- C) Anthracite
- D) Peat

Plant Substrate → Coal

Surface over which plant grows.

5. Which of the following is the oldest coalfield in India?

- A) Jharia → Jharkhand - Largest
- B) Bokaro
- ~~C)~~ Raniganj - West Benga
- D) Talcher


Crude oil

Petroleum

- The word 'Petroleum' has been derived from the two words 'petra' which means rock and 'oleum' which means oil.
- Thus, petroleum means 'rock oil'.
- It is a complex mixture of hydrocarbon compounds.
- There is not even a tiniest part of crude petroleum which goes waste or remains unused.

Petroleum

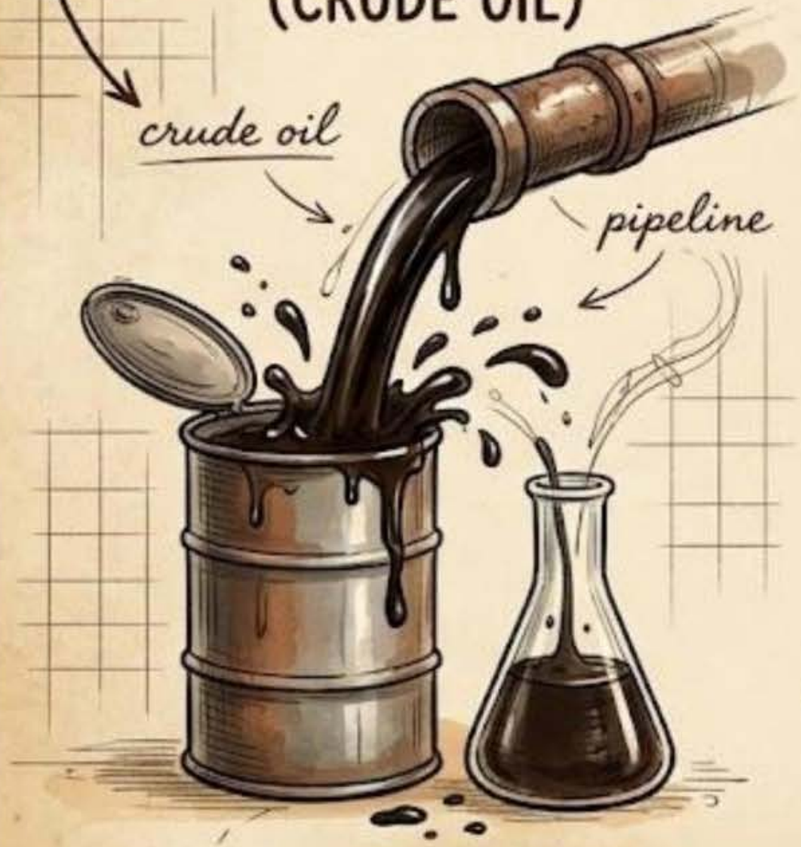
- **That is why petroleum is called 'liquid gold.'**

- The liquid petroleum is called crude oil,
- petroleum gas is called natural gas and
- the semi-solid to solid forms of petroleum are known as asphalt, tar, pitch, bitumen, etc.

STATES AND FORMS OF PETROLEUM

Hands made trende statcins, Your them are dor-ta by the of petroleum

LIQUID PETROLEUM (CRUDE OIL)



PETROLEUM GAS (NATURAL GAS)



SEMI-SOLID TO SOLID FORMS (ASPHALT, TAR, PITCH, BITUMEN)



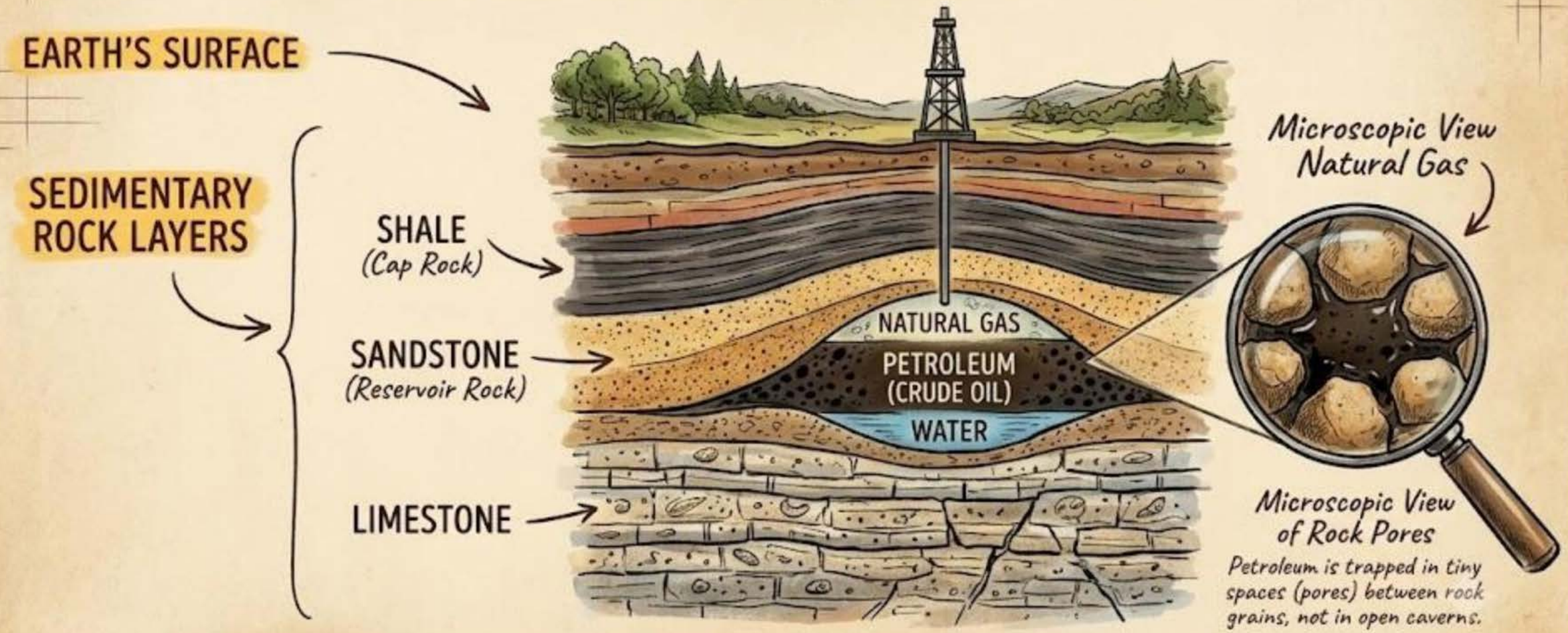
From liquid crude oil to natural gas and solid bitumen, petroleum naturally exists in various states.

Petroleum

- Petroleum is found in underground reservoirs in sedimentary rock formations like sandstone, shale and limestone.
- In some wells, crude oil is found along with natural gas and water.

WHERE PETROLEUM IS FOUND UNDERGROUND

Reservoirs in Sedimentary Rock Formations



Petroleum is found in underground reservoirs in sedimentary rock formations like sandstone, shale and limestone.

Advantages

- When drilled from the earth crude petroleum is a viscous dark liquid made up of many hydrocarbons.
- It is used both as fuel and raw material. In liquid form it can be transported through pipes or vehicles.
- Petroleum has high density; one kilogram of oil can generate 10,000 kcal of energy.
- It has broad area for applications as a source of energy.

Advantages

- Fuel: petrol, diesel, kerosene, jet fuel, LPG , for road, rail, sea and air transport.
- Raw material (petrochemicals): synthetic rubber, synthetic fibre, plastics, PVC, polystyrene, paints, varnish, cosmetics, printing ink, fertilizers.
- Lubricants: grease and oils for machines.
- Power generation: limited use in India (near oil wells/refineries, diesel and gas turbine stations).

THE USES OF PETROLEUM

Versatile Applications in Everyday Life

FUEL

Road, Rail, Sea & Air Transport

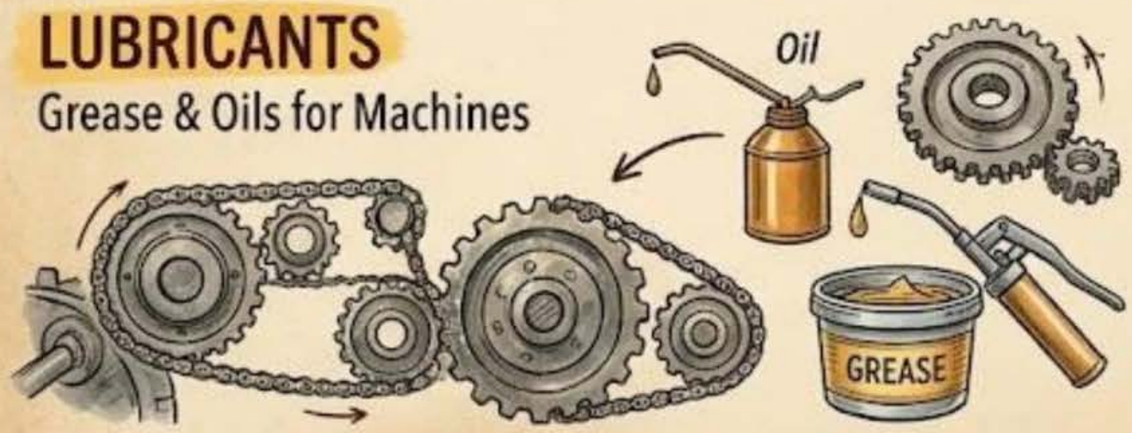


RAW MATERIAL (PETROCHEMICALS)



LUBRICANTS

Grease & Oils for Machines



POWER GENERATION

Limited Use in India

(Near oil wells/refineries, diesel & gas turbine stations)



Petroleum serves as a critical source of fuel, industrial raw materials, machine lubricants, and localized power generation.

Disadvantages

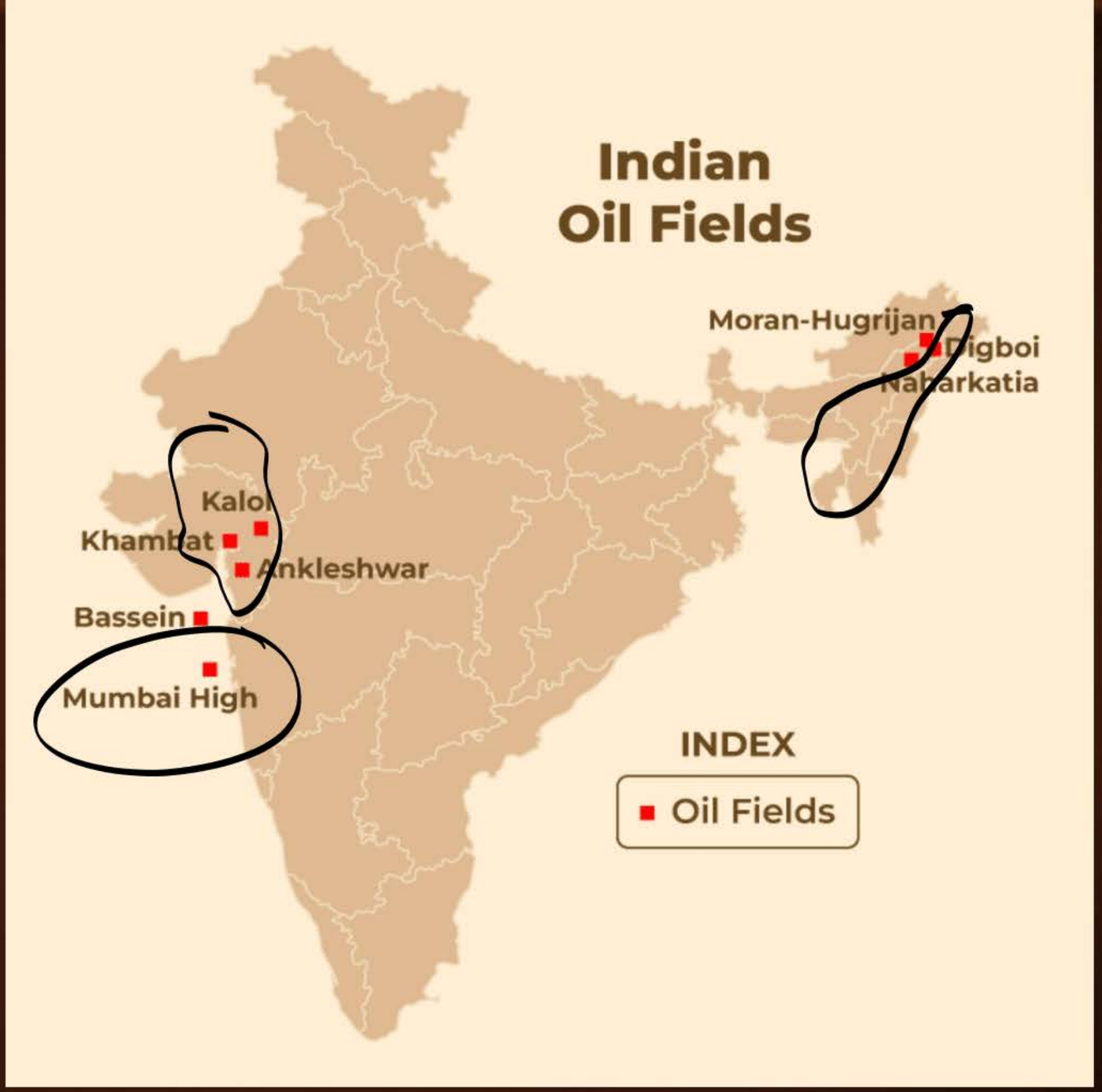
(i) Petroleum is a natural fossil fuel and non-renewable. So its availability is limited. With ever-increasing demand, petroleum resources are fast depleting हो रहा है

(ii) Extracting and burning petroleum generate Greenhouse Gases that contribute to environmental pollution and Global Warming.

(iii) Its costs are quite high because of its limited supply and high demand.

(iv) It is highly inflammable and can cause fires.

(v) During extraction from sea or transportation, if oil spills in water, it can pollute water and cause death of a number of marine animals.


Indian
Oil Fields
Moran-Hugrijan
Digboi
Naharkatia
Kalol
Khambat
Ankleshwar
Bassein
Mumbai High
INDEX
Oil Fields

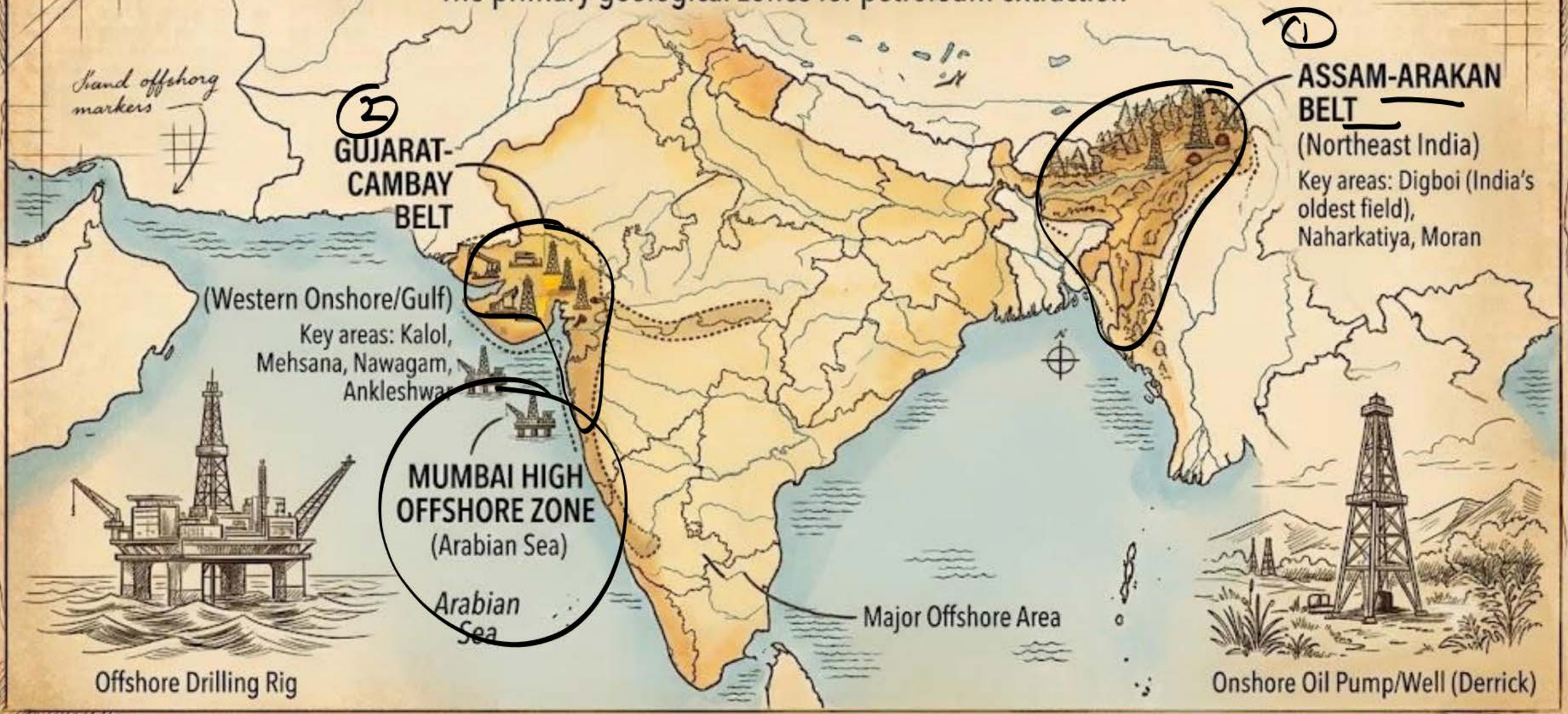
Distribution

- The entire oil production of India so far comes from
  1. the Assam-Arakan belt,
  2. the Gujarat-Cambay belt and
  3. the Mumbai High offshore zone.

1) ASSAM (north-east) : Digboi, Naharkatiya, Moran, Rudrasagar
2) GUJARAT (west) : Cambay basin, Ankleshwar, Kalol, Nawagam, Kosamba
3) MUMBAI HIGH (offshore, Arabian Sea): most productive field

THE THREE MAJOR OIL PRODUCING BELTS OF INDIA

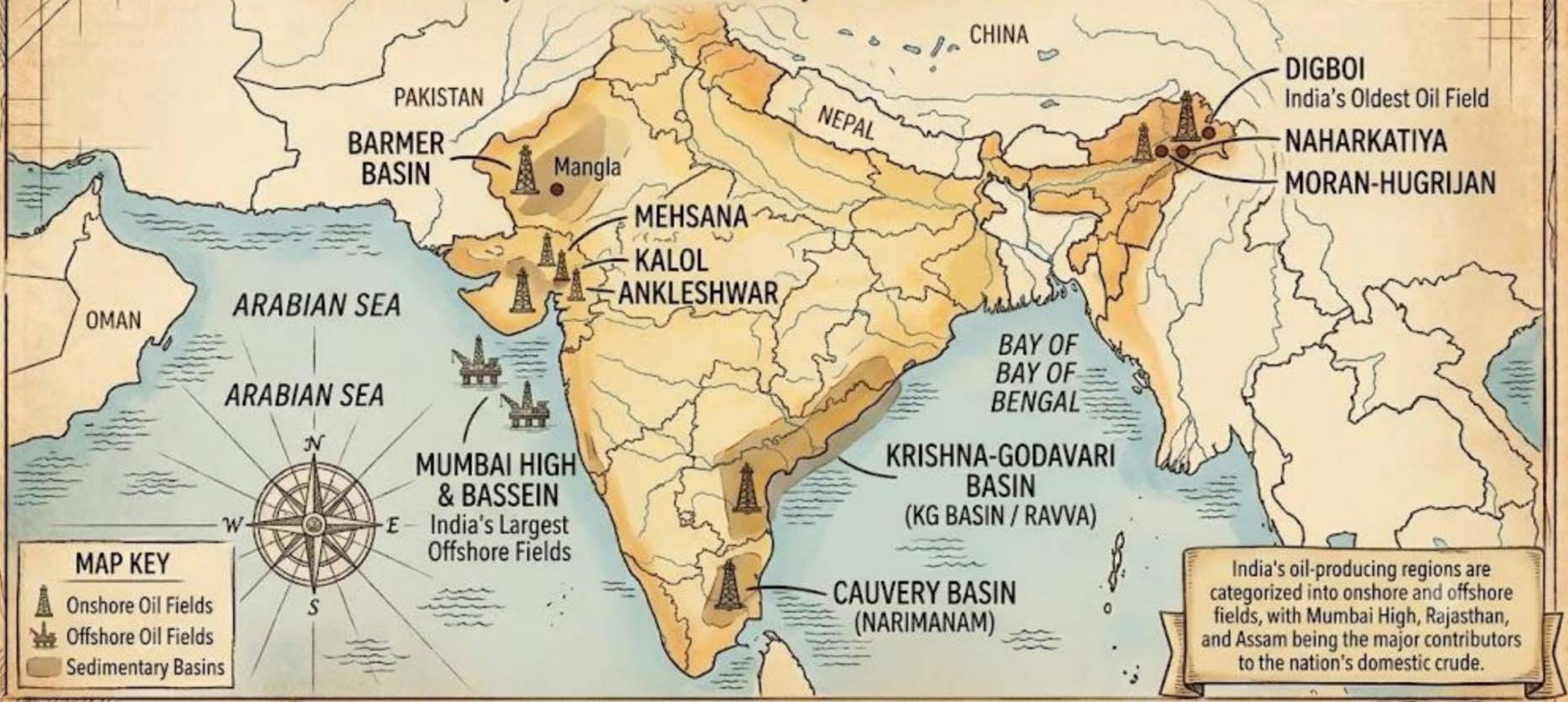
The primary geological zones for petroleum extraction



The entire production of India comes primarily from the Assam-Arakan belt, the Gujarat-Cambay belt, and the Mumbai High offshore zone.

MAJOR OIL FIELDS OF INDIA

Key Onshore & Offshore Hydrocarbon Reservoirs



The major oil fields are located in sedimentary basins, supporting the nation's fuel requirements.

In order of importance the main areas of mineral oil deposits are:

(1) **Mumbai High.** In Maharashtra, about 176km off Mumbai in the Arabian Sea, a huge oil deposit, known as Mumbai High, was struck in 1974.

- Mumbai High is the most productive oilfield and has a reserve of 5crore tonnes of oil.
- It is higher than the normal height and oil is taken from the depth of over 1,400m with the help of a specially designed platform known as 'Sagar Samrat'.

In order of importance the main areas of mineral oil deposits are:

(2) Oilfields of Eastern Region.

- India oldest and important oil bearing area in the eastern part of the Himalayas is Assam.
- **Digboi oilfield situated in Tinsukhia district of Assam is the oldest oilfield and the oldest oil refinery in India.**
- Moran, Bappapung, Hausanpung and Hugirijang also have oilfields.

In order of importance the main areas of mineral oil deposits are:

(3) Oilfields of Western Region.

- In Gujarat, Cambay basin is the site of the main oil bearing sand.
- The other oilfields of importance in Gujarat are Kalol, Koyali, Kosamba, Sanand, Kathana, Ankleshwar and Navgaon.
- Besides these oilfields, a number of potential oil bearing fields have been discovered in the States of Tripura, Punjab, Nagaland, Gujarat, West Bengal and Jammu and Kashmir and also in the Kaveri, the Krishna and the Godavari basins in the south.

11. Which is the oldest oil-producing state in India where petroleum was first discovered?

- A) Gujarat
- B) Maharashtra
- C) Assam - Digboi
- D) Tamil Nadu

16. Which of the following is an offshore oil field located in Gujarat?

- A) Ankleshwar
- B) Digboi
- C) Aliabet
- D) Naharkatiya

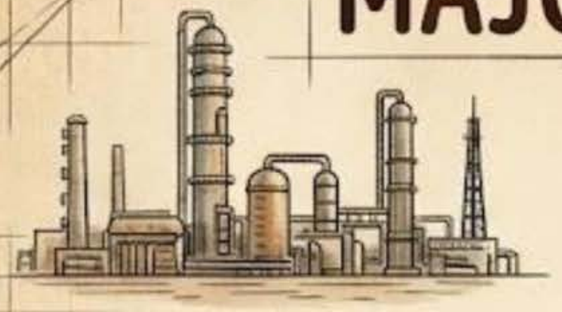
18. Why is petroleum preferred over coal for modern transportation?

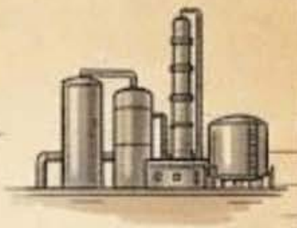
- A) It is heavier and provides vehicle stability
- B) It has a higher calorific value, produces little ash, and is easily transportable
- C) It is found evenly distributed across all Indian states
- D) It is a non-combustible fuel

Oil Refineries

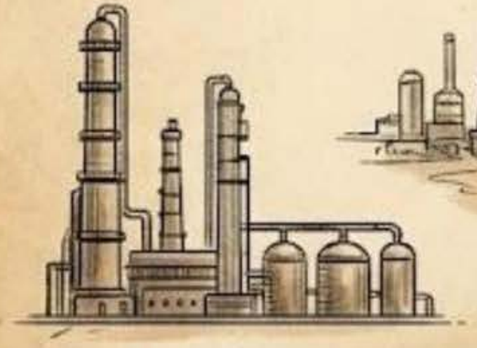
- Most of the refineries are located near the oilfield or near the coast to minimise the cost of transport and to avoid the risk of transporting the oil inside the country due to its inflammable nature.

MAJOR OIL REFINERIES OF INDIA

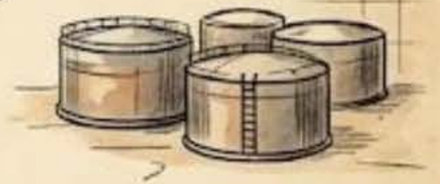












BATHINDA (Punjab)

MATHURA (Uttar Pradesh)

DIGBOI (Assam) - India's Oldest

PANIPAT (Haryana)

BARAUNI (Bihar)

GUWAHATI (Assam)

JAMNAGAR (Gujarat) - World's Largest

BONGAIGAON (Assam)

VADINAR (Gujarat)

BINA (Madhya Pradesh)

NUMALIGARH (Assam)

KOYALI / VADODARA (Gujarat)

HALDIA (West Bengal)

PARADIP (Odisha)

MUMBAI (Maharashtra)

VISAKHAPATNAM (Andhra Pradesh)

MANGALORE (Karnataka)

CHENNAI / MANALI (Tamil Nadu)

KOCHI (Kerala)

N

W

E

S

Legend / Key

Refinery Location

India has a vast network of refineries, with the world's largest complex in Jamnagar and historic operations dating back to Digboi in Assam.

Coal, Oilfields and Refineries in India



PYQ

1) What are the disadvantages of using Coal as a Source of power.

2) Mention any two Coalfields in India.

Thank You